

LUCIAN URITU

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PROFILE

Computer Science student at Maastricht University with hands-on experience in [software engineering](#), [high-performance computing](#), [networking](#), and [embedded systems](#). I have built projects ranging from [MPI-based traffic simulations](#) and [network protocols](#) to [eye-tracking applications](#), and I enjoy solving complex technical problems by turning ideas into working systems. I am looking to further develop as a [software engineer](#) while working on challenging, real-world problems.

EDUCATION

[Maastricht University](#), *Maastricht, The Netherlands*
Bachelor of Science in Computer Science

September 2024 - Expected July 2027

[IPLT 'Spiru Haret'](#) , *Chisinau, Republic of Moldova*
Baccalaureate Diploma (study profile 'Real')

September 2021 – July 2024

Exam Subjects: Mathematics, English, Romanian language and literature, Biology;
Subjects studied: Mathematics(advanced), English, Romanian language and literature,
Biology(advanced), Chemistry(advanced), Physics(advanced), Informatics(advanced), Romanian and
Universal History, Geography(advanced);
Average grade of 8.95 / 10.

WORK EXPERIENCE

[“Student assistant”](#), *Maastricht, The Netherlands*

April 2026 - Present

- Working as a student assistant at Maastricht University, [Open Science in Education team](#).
- Proposed an [original platform idea](#) to the team, which was [approved for development](#), independently [designed and developed the application end-to-end](#), with an [official launch planned for November 2026](#).
- Collaborate with [researchers and faculty](#) to improve the accessibility and usability of digital learning materials.
- Prepare [technical documentation](#) and user guides for students and staff.

PROJECTS

[“MUP Authoring Tool”](#), *Maastricht, The Netherlands*

April 2026- Present

- Conceived and proposed the idea for a new [Open Science educational platform](#) at Maastricht University Press. The proposal was approved by the team for development and deployment.
- Independently responsible for [the end-to-end development](#) of the platform, from initial concept and technical design to implementation and testing.
- Developed the platform using [JavaScript](#), [Node.js](#), [Electron](#), [HTML/CSS](#), [GitHub API](#), and [TeachBooks](#)

with a focus on helping non-technical authors convert plain text into structured Markdown books, preview content in real time, manage chapters and citations, and publish Open Science educational materials through GitHub Pages.

- Preparing the platform for its [official launch in November 2026](#), where it will be made available to [12 users pilot](#) Maastricht University.
- Project link: To be added after official launch.

[“Unblocking the City”](#), *Maastricht, The Netherlands*

January 2026 - June 2026

- Worked within a [6-person team](#) on a broad High-Performance Computing project, taking initiative in defining the specific research direction of [parallel urban traffic simulation](#) from the broader umbrella topic.
- Contributed to [project planning and task coordination](#), helping divide technical responsibilities, manage milestones and integrate individual components into the final simulation pipeline.
- Worked on writing the [MPI communication](#) between different ranks.
- Developed and evaluated parallel traffic simulations using [MPI](#), [SUMO](#), and [HPC clusters](#), exploring graph-partitioning approaches including METIS, Louvain, Spectral, and KD-based partitioning.
- Evaluated scalability through [speedup](#), [efficiency](#), and [bottleneck analysis](#), including parallel Monte Carlo experiments achieving up to [14.9x speedup on 32 CPU cores](#).
- Github link for the project: [LucianUritu/hpc-traffic-simulation](#)

[“Gemini-Lite Protocol”](#), *Maastricht, The Netherlands*

October 2025 - December 2025

- Developed a [lightweight Gemini protocol server and client](#) in java, featuring a [modular architecture](#) with support for [proxying](#), [file handling](#), and [custom protocol extensions](#).
- Implemented [robust networking](#), [request/response parsing](#), and integrated comprehensive testing.
- Explored network protocols through [practical analysis of DHCP traffic](#) and hands-on packet inspection.
- Github link for the project: [LucianUritu/GeminiLite-Protocol](#)

[“Parallax Effect User Study”](#), *Maastricht, The Netherlands*

September 2025 - January 2026

- A [game-based](#) platform designed to explore user interaction with parallax effects. In each level, the number of background particles changes to study how this variation impacts [user experience](#).
- The platform leverages [real-time eye tracking](#), [camera-based position estimation](#), [interactive 3D visualization](#) for immersive and dynamic gameplay.
- Github link for the project: [LucianUritu/Parallax-User-Study](#)

[“Helsinki Route Planner”](#), *Maastricht, The Netherlands*

January 2025 - June 2025

- A [JavaFX-based application](#) for planning [public transport routes](#) in Helsinki, featuring [real-time journey planning](#) and transit [heatmap visualization](#). The app allows users to plan routes from the chosen destination within the city. In addition to traditional route planning, it includes a heatmap feature that helps users visualize popular or high-traffic areas, enabling smarter travel decisions.
- The application was designed with a focus on user experience, real-world usability, and the specific urban mobility needs of Helsinki.
- Github link of the project: [GitHub - LucianUritu/Helsinki-Route-Planner](#)

[“Graph Coloring Game”](#), *Maastricht, The Netherlands*

September 2024 - December 2024

- This project is a [Graph Coloring Game](#), developed as part of a university assignment in collaboration with fellow students. It is an interactive [JavaFX](#) application that allows users to explore graph theory concepts through various graph coloring algorithms and game modes.
- Github link of the project: <https://github.com/LucianUritu/Graph-Coloring>

AWARDS

“Marathon C++ Tekwill”, Chişinău, Republic of Moldova

January 2023

- A national Informatics competition to solve 4 complex programming problems.
- 250/400 points
- Learned more how to [properly manage my time](#) and developed my knowledge in programming algorithms

“School Competitions”, Chişinău, Republic of Moldova

September 2021 - December 2023

- School Competitions at Informatics organized by the ministry of education every year.
- 10th Grade: II place, local competition; Mention, district competition
- 11th Grade: I place, local competition; Mention, district competition

LANGUAGES AND IT SKILLS

- [Romanian](#) (native), [English](#) (C2), [Russian](#) (conversational),
- Advanced user of [Microsoft Office](#) Excel, Word and PowerPoint, [Adobe Photoshop](#), [Premier Pro](#), [After Effects](#)
- [Programming Languages](#): Assembly, C, Java, Python, JavaScript, HTML/CSS
- [Technologies & Tools](#): Git, GitHub API, CUDA, MPI, OpenMP, Linux/Unix, SUMO, Node.js, OpenCV
- [Software Engineering](#): Object-Oriented Programming, Data Structures & Algorithms, Algorithmic Design, Databases, Web Development, Network Programming, API Integration,
- [Systems & Computing](#): High-Performance Computing, Embedded Programming, Parallel & Distributed Computing, Computer Networks, IoT, Blockchains.